

Barro-Becker with Fertility Inertia^{*}

HongSeok Kim[†]

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Abstract

Why do large pronatalist policies often produce small fertility responses? This paper develops a dynastic fertility model in which parents observe public news about future childrearing conditions but are uncertain about how informative that news is for their own household. Parents choose fertility and child investment under a cautious interpretation of the signal. The model delivers fertility inertia: over a range of moderately adverse signals, news lowers welfare but leaves fertility, child investment, and consumption unchanged. The mechanism is that bad news both lowers the expected value of children and, if informative, resolves uncertainty. A cautious parent adjusts the interpretation of the signal until these forces offset at the margin. The theory predicts muted responses to good news, sharp responses to severe bad news, and inaction after moderately bad news. It implies that pronatalist policy should be evaluated not only by fiscal generosity but also by legibility. Policies that make household-specific parenthood payoffs clearer can have larger effects than larger but general transfers.

Keywords: Fertility; pronatalist policy; ambiguity aversion; information quality; fertility inertia; dynastic choice; child investment.

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[†]Iowa State University. Email: hongseok@iastate.edu. Website: <https://hongseokimm.github.io/>.

1 Introduction

Since 2006, South Korea has spent roughly 380 trillion won, which is about 270 billion U.S. dollars, on policies to raise its birth rate: cash grants at birth, monthly child allowances, subsidized housing for newlyweds, paid parental leave, expanded childcare, and fertility-treatment subsidies. Over the same period its total fertility rate fell from 1.13 to 0.72, declining in nearly every program year. In Seoul it reached 0.55, the lowest recorded for any national jurisdiction. Japan, which calls demographic decline its central long-run challenge and now commits 3.5 trillion yen annually to family policy, recorded fewer than 700,000 births in 2024 and has posted record low fertility for nine straight years. These transfers are huge, yet the fertility response has been small. A further pattern compounds the puzzle. Across low-fertility countries, stated ideal family size has stayed near two children even as realized fertility has fallen (Sobotka and Beaujouan 2014). Households report wanting children and face governments offering substantial financial resources to have them, and yet appear not to adjust when incentives are provided.

The standard economic model of fertility offers limited understanding on these observations. In the dynastic framework of Barro and Becker (1989) and Becker and Barro (1988), parents choose fertility by equating the marginal cost of an additional child to its marginal dynastic benefit, so that a transfer lowering the cost or raising the benefit should induce a measurable response. The possible explanations, such as inattention, adjustment costs, slow norm transmission, are hard to reconcile with the salience of these programs.

This paper develops an alternative explanation. We argue that the weak fertility response to public news and pronatalist policy need not reflect inattention, irrationality, or unusually large adjustment costs. Instead, households may understand the information they receive but remain unsure about how strongly it applies to their own future. A national childcare announcement, a housing policy, or a labor-market forecast may be clearly relevant for parenthood in general. But its relevance for a particular household with its own location, occupation, eligibility status, family support, and timing is often much less clear. Fertility may therefore remain unchanged not because households ignore information, but because they cannot confidently convert public information into a private parenthood payoff.

The paper formalizes this idea based on a dynastic theory of fertility. Parents care about their own consumption and about the future well-being of their children. They choose how many children to have and how much to invest in them. Before making that choice, they observe public news about the future environment their children may face. The key assumption is that parents are unsure how informative this news is for their own dynasty. A cautious parent evaluates a family plan under the least favorable interpretation of the news.

The central mechanism has two parts. Bad news about the future lowers the expected value of having children. At the same time, informative news also resolves uncertainty. For a risk-averse parent, resolution of uncertainty has value, especially because children in the same family are exposed to many of the same future conditions, such as the same housing market, labor market, school system, childcare regime, and policy environment. A larger family is not a diversified portfolio of independent outcomes. It is a larger exposure to a shared future risk.

These two forces work in opposite directions after bad news. If the news is taken as highly informative, it makes the bad news more damaging. But it also makes the future less uncertain. If the news is taken as weakly informative, the bad news is less damaging, but more uncertainty remains. A cautious parent chooses the interpretation that is worst for the family plan under consideration. For moderately bad news, that worst-case interpretation adjusts in exactly the way that offsets the effect of the news on the marginal value of fertility.

This produces the main result: *fertility inertia*. There is a range of moderately adverse public signals over which the parent's welfare changes but her choices do not. Fertility, child investment, and consumption remain fixed even though the parent observes and processes the news. The news matters for how the parent feels about the future, but it does not change the marginal tradeoffs that determine the family plan. In this sense, the household is not inattentive. The response occurs in beliefs rather than in observed fertility.

The result also explains why the response to news is asymmetric. Good news is discounted because a pessimistic parent treats favorable information as only weakly applicable to her own family. Very bad news, by contrast, is taken seriously because the worst-case interpretation is that the bad news is highly informative. Between these cases lies the inertia region, where the interpretation of the signal adjusts and the fertility choice is flat. The model therefore predicts muted responses to good news, sharp responses to severe bad news, and no response to moderately bad news.

This mechanism requires both ambiguity and risk aversion. Ambiguity alone makes parents pessimistic about how to read the news, but it does not produce a flat region of fertility choices. Risk aversion alone makes parents dislike unresolved uncertainty, but if the meaning of the news is known, fertility still responds smoothly to the signal. Inertia appears only when parents are both unsure how to interpret the news and concerned about the common risk faced by their future children.

The theory delivers a simple comparative-static message. The no-response region is wider when parents are more sensitive to risk, when the family plan exposes more dynastic value to the common future environment, and when households are more uncertain about what public news means for their own circumstances. In the closed-form illustration, the width of the inertia

region is governed by two intuitive forces: how attractive parenthood is relative to its risk, and how ambiguous the signal is. More attractive parenthood can therefore be associated with a wider inertia region, because the parent has more at stake and is more exposed to the common future risk.

The policy implication is that pronatalist policy should distinguish between generosity and legibility. A larger transfer can raise the average payoff from having children, but it may not reduce uncertainty about whether the policy will actually apply to a household’s own future. By contrast, policies that make parenthood payoffs clearer can directly reduce the ambiguity that generates inertia. Examples include guaranteed childcare slots rather than lotteries, stable eligibility rules, credible long-horizon benefit commitments, enforceable parental-leave and job-return rights, predictable housing support, and clear caps on future child-related costs.

This distinction changes how weak fertility responses should be interpreted. A small response to a policy announcement does not necessarily mean that households did not notice the policy or that money does not matter. It may mean that the policy was not legible enough: households could not determine how it would change their own future parenthood payoff. The model predicts that fertility responses should be larger when policies are specific, credible, durable, and easy to map into household-level circumstances, even if those policies are not the largest in fiscal terms.

Related Literature The paper contributes to three literatures. First, it extends the economic theory of fertility. The modern theory begins with the quantityquality tradeoff of [Becker and Lewis \(1973\)](#) and the first dynamic optimizing treatment of fertility in [Razin and Ben-Zion \(1975\)](#), and is given its dynastic form in the Barro-Becker framework, which emphasizes the tradeoff between the cost of children and their discounted dynastic benefit ([Barro and Becker 1989](#); [Becker and Barro 1988](#)). A large subsequent literature embeds endogenous fertility and the quantity-quality margin in models of growth, human capital, and inequality ([Galor and Weil 2000](#); [de la Croix and Doepke 2003](#); [Doepke and Zilibotti 2005](#); [Manuelli and Seshadri 2009](#)). A parallel strand studies how fertility responds to income risk, employment risk, fecundity risk, and incomplete markets ([Ranjan 1999](#); [Sommer 2016](#); [Doepke et al. 2023](#)). In all of these models, households face risk under a known probability model. This paper studies a different problem. Households are unsure about the meaning of the information they receive. The uncertainty is not only about what the future will be, but about how strongly public news maps into the household’s own future.

Second, the paper’s decision-theoretic apparatus draws on the literature on choice under ambiguity. The maxmin expected-utility model of [Gilboa and Schmeidler \(1989\)](#) represents a cautious decision maker who evaluates each act under the least favorable of a set of priors;

Epstein and Schneider (2003) provide the dynamically consistent recursive extension that makes the criterion usable in a sequential problem; and the smooth-ambiguity model of Klibanoff et al. (2005, 2009) separates ambiguity from the attitude toward it. The robustness program of Hansen and Sargent (2001, 2008) recasts the same concern as a desire for decisions that perform well across a set of models surrounding a benchmark. Our parent is a maxmin decision maker of this kind, evaluating each family plan under the least favorable admissible interpretation of public news, and the worst-case measure that prices her continuation value is the object these foundations supply.

Third, the result is most closely related to a strand of finance that studies ambiguity about the quality of information rather than about payoffs, and the inertia it produces. Epstein and Schneider (2008) show that investors uncertain about signal precision evaluate news under a worst-case assessment of its quality and therefore react more strongly to bad news than to good news. Caskey (2009) shows that ambiguity-averse investors may prefer coarse, aggregated signals, so that prices fail to impound public information, while Ju and Miao (2012) embed ambiguous learning in a dynamic asset-pricing model. Closest to our setting, Illeditsch et al. (2021) establish information inertia in which the combination of risk and ambiguity makes optimal portfolios unresponsive to news worse than expected, an effect that vanishes when investors are risk-neutral or the news concerns purely idiosyncratic risk, and that rationalizes low household trading. We apply this logic from financial markets to the family. The relevant “asset” is a child which is a long-term, non-traded, non-diversifiable claim on the family’s future. Because the claim cannot be sold or hedged, the inertia margin is a forgone birth rather than a forgone trade. Because the continuation value is endogenous, it is the next generation’s dynastic value rather than an exogenous payoff, and because descendants share a common future environment, a larger family is a more concentrated position in a single undiversifiable risk. This is the opposite of the idiosyncratic case in which financial information inertia disappears. To our knowledge, the combination of information-quality ambiguity with an endogenous dynastic continuation value is new to both literatures.

Outline The remainder of the paper proceeds as follows. Section 2 develops the recursive Barro–Becker model with ambiguous public signals, establishes existence and uniqueness, and derives the general optimality conditions. Section 3 uses a local normal representation to characterize signal informativeness, derive the fertility information inertia result, and provide the closed-form and quantitative illustrations. The final section 4 discusses empirical and policy implications.

2 A Recursive Barro–Becker Model with Ambiguous Signals

This section develops a recursive Barro–Becker model of fertility in which a parent observes public news about the parenthood environment her children will face but is unsure how relevant that news is for her own dynasty. The model delivers one sharp result: when the parent is averse to this kind of uncertainty – uncertainty about the informativeness of information itself – an entire interval of moderately bad public news leaves her fertility, her investment in each child, and her consumption completely unchanged. I refer to this as *fertility inertia*.

Informative bad news about parenthood acts through two opposing channels. It lowers the expected payoff to having children (the mean channel). At the same time, it resolves uncertainty about that payoff (the risk channel) and a risk-averse parent values this resolution because her children will all share the same future – the same labor market and policy regime – so the family carries a concentrated exposure to that future rather than a diversified one. A parent who does not know how informative the news truly is, and who evaluates her plans under the worst plausible interpretation, arrives at exactly the interpretation under which these two channels offset at the margin. Because the worst-case belief re-adjusts for each family plan, the offset holds at every plan simultaneously. The news lowers her welfare, but it moves none of her decisions.

Three building blocks are assembled. The first is the Barro–Becker dynasty. In their dynastic-altruism framework, a parent values children because she cares about their lifetime utility, those children in turn care about their children, and so on. The value of having a child today is therefore the discounted value function of the next generation, an endogenous object that must be solved for as a fixed point. The second is risk sensitivity. The next generation’s value is risky, since it depends on shocks that have not yet been realized, and the parent evaluates this risky continuation value through an exponential (CARA) certainty-equivalent operator. The same risk-sensitive recursion is used in the robustness literature following Hansen and Sargent (2001). The third is ambiguity about signal informativeness. The parent observes a public signal about the environment her children will face but does not know how informative that signal is for her own dynasty, and she guards against the worst plausible informativeness, in the maximin tradition of Gilboa and Schmeidler (1989).

The argument proceeds in three layers, from general to concrete. We first set up the recursive problem and establish that it is well posed (Proposition 1), then derive general first-order conditions, which take the form of standard Barro–Becker optimality conditions evaluated under a pessimistically distorted probability measure (Proposition 2). Second, a local normal approximation to the continuation value delivers the sharp results: a non-monotone effect of signal

informativeness on fertility (Proposition 3) and an interval of public news over which fertility, child investment, and consumption are all completely unresponsive (Proposition 4). This normal layer also nests, as an explicit special case, a transparent two-period model in which the payoff to a child is an exogenous random variable (Remark 1)—a useful mental model that contains the entire mechanism in miniature. Third, a closed-form special case expresses the width of the inertia region in terms of a “reproductive Sharpe ratio” and is used to compute the figures (Figures 1–3) that illustrate every result quantitatively.

2.1 Environment

The state. Time is discrete and each period corresponds to one generation. A parent enters the period with state

$$(x_t, s_t) \in \mathcal{X} \times \mathcal{S}.$$

The vector x_t collects everything payoff-relevant about the household’s own circumstances and the economy: assets, human capital, age, aggregate conditions, and policy variables. The scalar s_t is a public signal about the parenthood environment the next generation will face. For instance, news about housing costs, childcare availability, labor-market conditions, or announced family policy could be examples. High s_t is good news for future parenthood, while low s_t is bad news. Current resources are

$$m_t \equiv y_t + a_t,$$

where y_t is current income and a_t is assets, both functions of x_t .

The controls. The parent chooses consumption $c_t \geq 0$, fertility $n_t \geq 0$, and an investment or transfer per child $z_t \geq 0$, resources devoted to each child’s education, health, or bequest, the “quality” margin of the classic quantity–quality tradeoff. Fertility is treated as a continuous variable for tractability or can interpret fractional values of n_t as the intensity of the family plan (the probability or timing of an additional birth). The budget constraint is

$$c_t + n_t(q_t + z_t) \leq m_t, \tag{1}$$

where $q_t > 0$ is the direct resource cost of bearing one child, so each child costs its parent $q_t + z_t$ in total.

The transition and the ambiguous parameter. The next generation's state is generated by

$$(x_{t+1}, s_{t+1}) = G(x_t, s_t, z_t, \varepsilon_{t+1}; \rho), \quad (2)$$

where ε_{t+1} is a stochastic innovation realized after the parent's choices, with a distribution that does not depend on the controls. The parameter ρ indexes the family of probability models the parent perceives: it measures how informative the public signal s_t is for the next generation's continuation value. The parent knows the signal is relevant, but is ambiguous about how relevant:

$$\rho \in [\underline{\rho}, \bar{\rho}], \quad 0 < \underline{\rho} < \bar{\rho} < 1, \quad (3)$$

with no prior probability distribution over the interval. The interpretation is the natural one. For example, national headlines about housing costs or childcare shortages are clearly informative about parenthood in general, but a given household cannot know how strongly they map into the prospects of its own descendants, who will live in a particular place, work in particular occupations, and face particular schools and housing markets. The parent knows the signal is relevant ($\underline{\rho} > 0$) but not how relevant. This is Knightian uncertainty or ambiguity about signal informativeness, as distinct from ordinary risk about the shock realization itself. Expectations under a given model ρ and investment choice z_t are denoted

$$\mathbb{E}_{\rho, z_t}[\cdot \mid x_t, s_t] \equiv \mathbb{E}[\cdot \mid x_t, s_t]$$

which the subscripts are suppressed for simplicity.

Timing. Within each generation, the sequence of events is as follows: (1) the parent observes the public signal s_t , (2) the parent chooses consumption c_t , fertility n_t , and investment z_t , (3) the innovation ε_{t+1} realizes and the next generation's state and hence its continuation value is determined. Choices are therefore made under risk over ε_{t+1} and under ambiguity over ρ .

The dynastic objective. Let

$$V' \equiv V(x_{t+1}, s_{t+1})$$

denote the child's continuation value, or the child's own lifetime value, inclusive of her future fertility choices, evaluated by the same value function we are solving for. This recursive structure is the defining feature of the Barro–Becker framework. The return on a child is not a primitive payoff but the fixed point of the dynasty's own problem. Fertility scales the continuation value

through the Barro–Becker endogenous discounting factor $\Phi(n_t)$, where

$$\Phi(0) = 0, \quad \Phi'(n) > 0, \quad \Phi''(n) \leq 0.$$

A childless dynasty receives no continuation value. Additional children always add dynastic weight, and the marginal weight is diminishing. The second child contributes less additional altruistic value than the first. The standard parametric example is

$$\Phi(n) = \frac{n^{1-\eta}}{1-\eta}, \quad \eta \in [0, 1),$$

where η governs the curvature of dynastic altruism and $\eta = 0$ gives the linear case $\Phi(n) = n$. Note that none of the characterizations below requires this functional form.

2.2 Risk-sensitive continuation value

The continuation value V' is risky at the time the parent decides. We need a way for the parent to convert that risky prospect into a certainty-equivalent number she can trade off against current consumption. The model uses the exponential certainty-equivalent operator. That is, for any risky continuation payoff Y , define

$$\mathcal{C}(Y \mid x_t, s_t) = \mathcal{C}_{\gamma, \rho, z_t}(Y \mid x_t, s_t) \equiv -\frac{1}{\gamma} \log \mathbb{E}[\exp(-\gamma Y) \mid x_t, s_t], \quad \gamma > 0. \quad (4)$$

This operator returns the certainty equivalent of the gamble Y : the sure amount a CARA agent with absolute risk aversion γ would accept in exchange for facing a random variable Y . Here γ plays the role of an effective dynastic risk aversion. A larger γ penalizes variability more heavily, so the certainty equivalent sits further below the mean. As $\gamma \rightarrow 0$, the penalty vanishes and the operator reduces to the ordinary expectation $\mathbb{E}[Y]$. The same operator appears throughout the risk-sensitive control and robustness literature ([Hansen and Sargent 1995](#); [Tallarini 2000](#)), where it is known to encode an aversion to outcomes in the left tail. The following lemma records its key properties and, in particular, the exact mean–variance form it takes for normal payoffs—the workhorse formula of this section.

Lemma 1 (CARA certainty equivalent). *Let $U(Y) = -\exp(-\gamma Y)$. The sure payoff $\bar{Y}$ that gives the same utility as the risky payoff Y under probability model ρ satisfies*

$$U(\bar{Y}) = \mathbb{E}_{\rho, z_t}[U(Y) \mid x_t, s_t],$$

and solving for $\bar{Y}$ gives

$$\bar{Y} = -\frac{1}{\gamma} \log \mathbb{E}_{\rho, z_t} [\exp(-\gamma Y) \mid x_t, s_t] = \mathcal{C}_{\gamma, \rho, z_t}(Y \mid x_t, s_t).$$

If $Y \mid x_t, s_t, \rho \sim N(\mu_Y, \sigma_Y^2)$, the formula is exact and linear-quadratic,

$$\mathcal{C}(Y \mid x_t, s_t) = \mu_Y - \frac{\gamma}{2} \sigma_Y^2.$$

For general Y , a small- γ expansion gives

$$\mathcal{C}(Y) = \mathbb{E}[Y] - \frac{\gamma}{2} \text{Var}(Y) + o(\gamma),$$

and therefore

$$\lim_{\gamma \rightarrow 0} \mathcal{C}(Y) = \mathbb{E}[Y].$$

The derivation of CARA certainty-equivalent is provided in Appendix A.1. The normal case says that a risk-sensitive parent evaluates a normal prospect by its mean minus a variance penalty priced at $\gamma/2$. The expansion says that for small risk aversion, every prospect is evaluated this way. Both facts are used repeatedly below.

The recursive problem. The payoff to fertility is the scaled continuation value

$$Y_{t+1}(n_t, z_t) = \Phi(n_t) V(x_{t+1}, s_{t+1}). \quad (5)$$

The parent evaluates this payoff twice over. First, she applies the risk-sensitive operator within each probability model. Second, she then takes the worst case across models, in the max-min tradition of Gilboa and Schmeidler (1989) in which an ambiguity-averse agent evaluates each action under the worst probability model in her set of plausible models, and then chooses the action whose worst-case evaluation is best. Behaviorally, this captures a parent who, unable to assign odds to how relevant the news is for her, plans against the least favorable interpretation. The recursive problem is

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \mathcal{C}_{\gamma, \rho, z_t} [\Phi(n_t) V(x_{t+1}, s_{t+1}) \mid x_t, s_t] \right\} \quad (6)$$

subject to (1), where u is the parent's flow utility over own consumption ($u' > 0$, $u'' < 0$) and $\beta \in (0, 1)$ is the discount factor. The standard Barro–Becker fixed point is fully preserved. The value of a child is still the next generation's value function. What changes is only the operator

applied to that continuation value: a CARA certainty equivalent, taken first, with a minimization over interpretations of the news taken around it. The order is deliberate. The worst-case ρ is selected separately for each candidate plan (n_t, z_t) , so the worst-case interpretation of the news can shift as the plan changes. This plan-by-plan re-selection of the worst case is what drives the inertia result.

Interpretation To see why the order matters, it is worth to ask what each operator is doing. The inner certainty equivalent prices the risk the parent cannot avoid. Even once she has fixed an interpretation of the news, the next generation's payoff remains uncertain, and because her children all live through the same future, that residual risk is concentrated rather than diversified. The outer minimization prices a different concern. She does not know how informative the news is for her own dynasty, and rather than commit to a single guess about that informativeness, she evaluates each plan under the worst-case scenario.

Reversing the order would change the result entirely. If the parent fixed a single worst-case interpretation before choosing her plan, that interpretation would behave like a known parameter. A fixed feature of the environment would then pass through to every margin of the decision. The inertia depends on the opposite timing. It is because the worst case is reconsidered for each plan, rather than settled in advance, that the news can fail to move the decision at all.

The mechanism is easiest to see by following a parent who is deciding between a larger family and a smaller one. A larger family is a bigger undiversified bet on the common future, so it has more to lose if that future turns out badly. For such a plan, the interpretation the parent insures against is one that treats the news as highly informative, because informative bad news is exactly what a large family has most reason to fear. A smaller family is less exposed, so the worst case attached to it is less severe. The pessimism, in other words, is tailored to the plan under consideration. It bites hardest where the parent has the most at stake.

The two movements then track each other. As the parent considers scaling her family up, her exposure rises, but so does the informativeness she must brace for. As she considers scaling down, both fall together. Over a range of moderately bad signals these movements stay closely aligned enough that no plan looks strictly better than the one she would have chosen had the news never arrived.

This is why belief endogeneity is the engine of the result. Suppose instead that the parent held a single pessimistic belief that did not change with her plan. News operates through two channels. It lowers the expected payoff of children, and it resolves risk about that payoff. Under a fixed belief, these two channels would offset at exactly one plan. The policy function would then have a single kink at that plan and would respond to the news everywhere else. Because

the worst-case belief instead adjusts as the plan varies, the offset is no longer tied to a single plan. It holds across a whole interval of plans at once. The single kink is therefore replaced by a flat region. The parent still processes the news fully at every plan she considers. The news is neutralized because the worst-case interpretation she insures against is not fixed. It moves with each plan she considers, adjusting to keep that plan as unattractive as the others.

2.3 Recursive problem and solution concept

Because the value function appears on both sides of (6), we must verify that a solution exists and is unique. Define the Bellman operator T by

$$(TV)(x, s) = \max_{c, n, z} \left\{ u(c) + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \mathcal{C}_{\gamma, \rho, z} [\Phi(n) V(x', s') \mid x, s] \right\} \quad (7)$$

subject to

$$c + n(q(x, s) + z) \leq m(x, s).$$

A solution is a fixed point $V = TV$ together with policy functions

$$c^*(x, s), \quad n^*(x, s), \quad z^*(x, s),$$

and a worst-case belief policy

$$\rho^*(x, s) \in [\underline{\rho}, \bar{\rho}].$$

Assumption 1 (Regularity). *The feasible set is compact and nonempty for each (x, s) and varies continuously with the state; u is continuous, increasing, and strictly concave; and Φ is continuous and increasing. The operator T acts on the Banach space of bounded continuous functions on $\mathcal{X} \times \mathcal{S}$ equipped with the sup norm. Let*

$$\bar{\Phi} \equiv \sup_{n \in \Gamma_n} \Phi(n) < \infty,$$

where Γ_n is the (compact) set of feasible fertility levels, and assume $\beta \bar{\Phi} < 1$.

Assumption 1 draws on [Epstein and Schneider \(2003\)](#). Two features specific to our setting deserve comment. The first is that the operator T embeds a worst-case minimization over interpretations inside the dynastic recursion, and it is not automatic that this remains well behaved once iterated. Applied naively, a sequence of period-by-period worst-case choices can fail to be dynamically consistent. The interpretation a parent would insure against looking ahead from today need not coincide with the one she would insure against tomorrow, and the recursion can

then unravel. What rules this out is rectangularity of the set of interpretations together with prior-by-prior updating, the conditions [Epstein and Schneider \(2003\)](#) identify as necessary and sufficient for the recursive max-min operator to be dynamically consistent. Under these conditions, the order in which the parent solves the problem does not matter. The plan she judges best today remains the one she judges best as the future unfolds, so the recursion is internally consistent. Solving plan-by-plan, with the worst case re-selected at each plan, is then exactly what the dynamic problem asks for. The plan-dependence of the worst-case belief, which drives our inertia result, is a novel property of the solution and not an artifact. Layering the CARA certainty equivalent inside T does not change this, since it sends bounded functions to bounded functions and preserves the ordering between them.

The second is the condition $\beta\bar{\Phi} < 1$, which is the recursive heart of Barro–Becker models. The effective one-generation discount factor is not β but $\beta\Phi(n)$: a parent with many children weights the future more heavily, since more of the dynasty’s value lives there. For the dynastic value to be finite, this endogenous discounting must stay below one uniformly over feasible plans, or a parent could generate unbounded value by loading weight onto an ever-growing future. The budget constraint bounds feasible n and so makes $\bar{\Phi}$ finite, but finiteness alone is not enough. The amplified factor could still exceed one at the largest affordable family. The assumption closes this by requiring that even the maximum feasible plan still discounts the future. This is the dynastic analogue of the familiar $\beta < 1$, and it is what secures the contraction in our setting.

Proposition 1 (Existence and characterization). *Under [Assumption 1](#), T is a contraction on the bounded continuous functions with modulus at most $\beta\bar{\Phi}$, so a unique $V = TV$ exists. It is attained by policy functions solving the outer maximum and a worst-case belief $\rho^*(x, s)$ solving the inner minimum.*

Proof. The proof is available in [Appendix A.2](#). □

The economic interpretation of the proposition is that ambiguity aversion and risk sensitivity are added at no cost to tractability. One might worry that wrapping the continuation value in a worst-case minimization and a risk-sensitive certainty equivalent would loose the recursive structure that makes dynastic models operative. The proposition shows it does not. Both layers are nonexpansive in the sup norm, so they pass through the continuation value without inflating the Lipschitz constant, and the worst-case operator inherits the contraction modulus $\beta\bar{\Phi}$ of the standard Barro–Becker model unchanged. The dynasty’s problem therefore remains a well-posed fixed point, with a unique value function that can be computed by value-function iteration exactly as in the expected-utility case. Robustness and risk sensitivity affect where the optimum lies, not whether it exists or can be computed.

2.4 Equilibrium characterization

We now characterize optimal behavior in the general model. The optimality conditions retain a standard Barro–Becker form, except that all expectations are taken under a pessimistically distorted probability measure that the parent’s own ambiguity-aversion generates.

For a given state (x_t, s_t) , define the model- ρ certainty equivalent of a plan,

$$K_\rho(n, z; x_t, s_t) \equiv \mathcal{C}_{\gamma, \rho, z}[\Phi(n)V(x_{t+1}, s_{t+1}) \mid x_t, s_t], \quad (8)$$

and its worst case across models,

$$J(n, z; x_t, s_t) \equiv \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n, z; x_t, s_t). \quad (9)$$

Since $u'(c) > 0$, the budget constraint binds at an interior solution, $c_t^* = m_t - n_t^*(q_t + z_t^*)$. Hence the reduced problem is

$$\max_{n, z} \{u(m_t - n(q_t + z)) + \beta J(n, z; x_t, s_t)\}. \quad (10)$$

Let

$$\rho_t^* \in \arg \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t)$$

be the worst-case signal-informativeness parameter at the optimum. For this ρ_t^* , define the risk-sensitive distorted probability measure Q_t^* which is the worst-case transition law $P_{\rho_t^*, z_t^*}$ reweighted state by state. The Radon–Nikodym derivative is

$$\frac{dQ_t^*}{dP_{\rho_t^*, z_t^*}} = \frac{\exp\{-\gamma\Phi(n_t^*)V(x_{t+1}, s_{t+1})\}}{\mathbb{E}_{\rho_t^*, z_t^*}[\exp\{-\gamma\Phi(n_t^*)V(x_{t+1}, s_{t+1})\} \mid x_t, s_t]}.$$

Hence, when $\gamma\Phi(n_t^*) > 0$, future states with lower continuation value receive larger likelihood weights. In particular, for two successor states ω_1 and ω_2 ,

$$\frac{(dQ_t^*/dP_{\rho_t^*, z_t^*})(\omega_1)}{(dQ_t^*/dP_{\rho_t^*, z_t^*})(\omega_2)} = \exp\{-\gamma\Phi(n_t^*)[V(\omega_1) - V(\omega_2)]\}.$$

Thus if $V(\omega_1) < V(\omega_2)$, then ω_1 receives relatively more weight under Q_t^* than under $P_{\rho_t^*, z_t^*}$. The distorted measure therefore shifts probability mass toward low-value future states and away from high-value future states. Holding the baseline law and continuation value fixed, a larger γ makes this tilt more aggressive. As $\gamma \rightarrow 0$, the likelihood ratio converges to one, and Q_t^* collapses back to the undistorted worst-case law $P_{\rho_t^*, z_t^*}$.

The parent's caution enters in two stages, corresponding to two distinct forms of pessimism. First, she chooses the model pessimistically by selecting the worst-case signal informativeness parameter ρ_t^* from the ambiguity set which is the ambiguity channel. Second, conditional on that worst-case model, the risk-sensitive aggregator evaluates continuation payoffs using the exponentially tilted measure Q_t^* . This is the risk-sensitivity channel. The two channels compound: the parent first evaluates choices under the least favorable admissible transition law, and then, within that law, assigns relatively greater weight to low-continuation-value successor states.

This is the twisted measure familiar from risk-sensitive control and robust control. Its usefulness is that it converts the marginal valuation of continuation payoffs into an expectation under a distorted probability measure. Equivalently, the CARA certainty equivalent admits the entropy representation

$$-\frac{1}{\gamma} \log \mathbb{E}_{P_{\rho_t^*, z_t^*}} \left[e^{-\gamma Y_{t+1}} \mid x_t, s_t \right] = \min_{Q \ll P_{\rho_t^*, z_t^*}} \left\{ \mathbb{E}_Q[Y_{t+1} \mid x_t, s_t] + \frac{1}{\gamma} D(Q \parallel P_{\rho_t^*, z_t^*}) \right\},$$

where $Y_{t+1} = \Phi(n_t^*)V(x_{t+1}, s_{t+1})$ and the measure Q is absolutely continuous with respect to $P_{\rho_t^*, z_t^*}$. The minimizer of this entropy-penalized problem is Q_t^* . Thus it gives a convenient way to describe the parent's behavior. It acts like a pessimistic belief that puts extra weight on bad future states. However, the parent is not simply maximizing expected utility under Q_t^* . The distorted measure comes from the risk-sensitive problem itself, and it changes when the chosen policy changes. The underlying problem still includes the entropy cost that disciplines how far Q_t^* can move away from the baseline worst-case law.

Assume that the transition admits the differentiable shock representation (2), with V differentiable along the relevant directions, and define the marginal effect of child investment on the child's continuation value by

$$D_z V_{t+1} \equiv \nabla_x V(x_{t+1}, s_{t+1}) \cdot G_z^x(x_t, s_t, z_t, \varepsilon_{t+1}; \rho_t^*) + V_s(x_{t+1}, s_{t+1}) G_z^s(x_t, s_t, z_t, \varepsilon_{t+1}; \rho_t^*), \quad (11)$$

where G^x and G^s denote the x and s components of the transition. In words, $D_z V_{t+1}$ is the realized marginal return to investing one more unit in each child depending on the realization of shock ε_{t+1} .

Proposition 2 (Interior first-order conditions). *Assume the transition admits the differentiable shock representation (2), with a shock transition, and that V is differentiable. Let $D_z V_{t+1}$ be defined by (11). Suppose (c_t^*, n_t^*, z_t^*) is an interior optimum, the inner minimizer ρ_t^* is unique,*

and differentiation under the expectation is valid. Then

$$\rho_t^* \in \arg \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t),$$

and, letting Q_t^* denote the twisted measure associated with (ρ_t^*, n_t^*, z_t^*) , the optimal policies satisfy

$$c_t^* = m_t - n_t^*(q_t + z_t^*), \quad (12)$$

$$u'(c_t^*)(q_t + z_t^*) = \beta \Phi'(n_t^*) \mathbb{E}_{Q_t^*} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t], \quad (13)$$

$$n_t^* u'(c_t^*) = \beta \Phi(n_t^*) \mathbb{E}_{Q_t^*} [D_z V_{t+1} \mid x_t, s_t]. \quad (14)$$

If $K_\rho(n_t^*, z_t^*; x_t, s_t)$ is differentiable in ρ , the worst-case belief also satisfies the one-dimensional optimality condition

$$\begin{cases} \partial_\rho K_\rho(n_t^*, z_t^*; x_t, s_t)|_{\rho=\rho_t^*} = 0, & \rho_t^* \in (\underline{\rho}, \bar{\rho}), \\ \partial_\rho^+ K_\rho(n_t^*, z_t^*; x_t, s_t)|_{\rho=\underline{\rho}} \geq 0, & \rho_t^* = \underline{\rho}, \\ \partial_\rho^- K_\rho(n_t^*, z_t^*; x_t, s_t)|_{\rho=\bar{\rho}} \leq 0, & \rho_t^* = \bar{\rho}. \end{cases} \quad (15)$$

Proof. The proof is available in Appendix A.3. $\square$

Proposition 2 shows that the parent's robust, ambiguity-averse problem delivers first-order conditions with the same basic structure as the standard Barro–Becker model. The household still equates marginal costs to marginal dynastic benefits at each margin. What changes is the way the forward-looking benefits are evaluated: the relevant marginal continuation terms are expectations under the pessimistically distorted measure Q_t^* , which is constructed from the worst-case signal-informativeness parameter ρ_t^* .

Equation (13) is the fertility margin. The left side is the marginal utility cost of an additional child. An extra child requires resources $q_t + z_t^*$: the fixed child cost q_t plus the per-child investment z_t^* . Valued at the marginal utility of consumption, this cost is

$$u'(c_t^*)(q_t + z_t^*).$$

The right side is the discounted marginal dynastic benefit of fertility. An additional child raises the dynastic weight by $\Phi'(n_t^*)$, and the continuation value attached to that child is evaluated under the distorted measure Q_t^* . Thus

$$\beta \Phi'(n_t^*) \mathbb{E}_{Q_t^*} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t]$$

is the pessimistically evaluated marginal value of an additional child. The fertility condition therefore says that a child is worth having until its marginal cost in forgone consumption equals its discounted dynastic value under the parent's worst-case, risk-sensitive evaluation.

Equation (14) is the child-investment, or quality, margin. Raising z_t by one unit costs one unit of resources for each of the n_t^* children, so the marginal utility cost is

$$n_t^* u'(c_t^*).$$

The benefit is the marginal effect of investment on the child's future state, valued through the continuation value function. This realized marginal return is $D_z V_{t+1}$. Since investment improves the continuation value attached to the existing children, the benefit is scaled by the total dynastic weight $\Phi(n_t^*)$ and averaged under Q_t^* . Hence the investment condition is

$$n_t^* u'(c_t^*) = \beta \Phi(n_t^*) \mathbb{E}_{Q_t^*} [D_z V_{t+1} \mid x_t, s_t].$$

Fertility is scaled by the marginal dynastic weight $\Phi'(n_t^*)$, because it adds another child to the dynasty. Child investment is scaled by the total dynastic weight $\Phi(n_t^*)$, because it improves the future state of the children already being chosen.

Dividing (13) by (14) eliminates the marginal utility of consumption and gives the quantity–quality condition

$$\frac{q_t + z_t^*}{n_t^*} = \frac{\Phi'(n_t^*)}{\Phi(n_t^*)} \frac{\mathbb{E}_{Q_t^*} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t]}{\mathbb{E}_{Q_t^*} [D_z V_{t+1} \mid x_t, s_t]}. \quad (16)$$

The left side is the marginal resource cost of fertility relative to the marginal resource cost of raising per-child investment. The right side is the marginal-to-total dynastic weight, $\Phi'(n_t^*)/\Phi(n_t^*)$, multiplied by the ratio of two pessimistic continuation terms: the level of continuation value carried by a child and the marginal return to investing in that child. This is the robust, signal-informativeness analogue of the Barro–Becker quantity–quality condition.

Equation (15) characterizes the worst-case belief. Given the parent's chosen plan, the inner minimization selects the signal-informativeness parameter ρ_t^* that delivers the lowest risk-sensitive continuation value. Thus ρ_t^* is the least favorable admissible interpretation of the news for that plan. If the minimizer is interior, then

$$\partial_\rho K_\rho(n_t^*, z_t^*; x_t, s_t) \big|_{\rho=\rho_t^*} = 0,$$

so small changes in signal informativeness have no first-order effect on the minimized objective. If the minimizer is at a boundary, the one-sided derivative has the sign required for a constrained

minimum: at $\underline{\rho}$, moving right weakly raises K_ρ , and at $\bar{\rho}$, moving left weakly raises K_ρ . Since larger ρ means more informative news, these cases correspond to maximally uninformative and maximally informative worst-case interpretations. This condition also governs how the worst-case belief changes with the parent's information. When the same ρ_t^* remains active over a region, the parent's response to news can be muted.

Because the same distorted measure Q_t^* appears in both the numerator and the denominator, one might expect the distortion to cancel out. In general it does not. The numerator is the level of continuation value, whereas the denominator is the marginal effect of investment on continuation value. Pessimism reweights future states toward those with low continuation value, but it need not reweight the level and the investment return in the same way. If low-value states are also states where investment has little payoff, the denominator may fall more than the numerator, raising the right side of (16). If low-value states are instead states where investment is especially productive, the denominator may rise relative to the numerator, lowering the right side. Thus ambiguity and risk sensitivity do not merely scale fertility up or down. They can also change the quantity–quality tradeoff by changing how the parent values the level of child continuation value relative to the marginal return to investment.

2.5 Risk-neutral and ambiguity-neutral limits

The model differs from the standard Barro–Becker problem in two ways. First, the parent is ambiguity averse: she evaluates each plan under the worst admissible signal-informativeness parameter $\rho \in [\underline{\rho}, \bar{\rho}]$. Second, she is risk sensitive: future continuation values enter through the CARA certainty equivalent rather than through a standard expectation. This subsection separates the two forces.

Start by shutting down risk sensitivity. By Lemma 1, for any bounded continuation payoff Y ,

$$\mathcal{C}_{\gamma, \rho, z}(Y) = \mathbb{E}_{\rho, z}[Y] - \frac{\gamma}{2} \text{Var}_{\rho, z}(Y) + o(\gamma).$$

Hence, as $\gamma \rightarrow 0$, the variance penalty disappears and the certainty equivalent converges to a conditional expectation. The Bellman equation therefore becomes

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \mathbb{E}_{\rho, z_t} [\Phi(n_t) V(x_{t+1}, s_{t+1}) \mid x_t, s_t] \right\} \quad (17)$$

which is still a robust problem. The parent is no longer risk sensitive, but she still insures against the least favorable value of ρ .

Because $\Phi(n_t)$ is chosen at time t and is nonnegative, it can be taken outside both the

expectation and the minimization. Thus

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \Phi(n_t) \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \mathbb{E}_{\rho, z_t} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t] \right\}. \quad (18)$$

Equation (18) is the risk-neutral robust Barro–Becker problem. Fertility still scales the value of descendants through $\Phi(n_t)$, but the continuation value is evaluated under the worst admissible signal-informativeness parameter.

Now shut down ambiguity. If the ambiguity set is a singleton, $\underline{\rho} = \bar{\rho} = \rho$, there is no longer an inner minimization. With risk sensitivity still present, the problem becomes

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \mathcal{C}_{\gamma, \rho, z_t} [\Phi(n_t) V(x_{t+1}, s_{t+1}) \mid x_t, s_t] \right\}. \quad (19)$$

This is ambiguity-neutral but still risk sensitive. The parent uses a single transition law, indexed by ρ , but still evaluates future outcomes through the CARA certainty equivalent.

Finally, if both features are switched off, that is $\gamma \rightarrow 0$ and $\underline{\rho} = \bar{\rho} = \rho$, the problem collapses to the standard Barro–Becker recursive formulation

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \Phi(n_t) \mathbb{E} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t] \right\}. \quad (20)$$

Thus the standard model is recovered only when the parent uses a single probability law and evaluates continuation value by an ordinary expectation.

3 Signal Informativeness and Fertility Inertia

This section isolates the mechanism. The full recursive model generally does not yield closed-form policy rules, so we use a local normal representation of the next generation’s continuation value. This representation preserves the economic meaning of the recursive model but makes transparent how signal informativeness affects fertility. The key result is fertility information inertia: over a range of moderately bad public signals, welfare changes but fertility, child investment, and consumption do not.

3.1 Local normal representation

The full recursive problem generally has no closed-form solution without more structure on preferences, fertility aggregation, and the transition law. To isolate the mechanism, we use a local normal approximation to the next generation’s continuation value. Around the current

state, suppose

$$V(x_{t+1}, s_{t+1}) = \mu(z_t) + \sigma(z_t) \left[\rho s_t + \sqrt{1 - \rho^2} \varepsilon_{t+1} \right], \quad \varepsilon_{t+1} \sim N(0, 1), \quad (21)$$

with $\sigma(z_t) > 0$.

This approximation should be read as a local representation of the endogenous Barro–Becker continuation value, not as an exogenous payoff. The functions $\mu(z_t)$ and $\sigma(z_t)$ summarize how parental investment changes the child’s future prospects. Investment may raise the expected continuation value, through $\mu(z_t)$, and it may also change the child’s exposure to the common economic environment, through $\sigma(z_t)$.

The signal-informativeness parameter ρ has two effects. Conditional on the current signal s_t ,

$$\mathbb{E}[V(x_{t+1}, s_{t+1}) \mid s_t, \rho] = \mu(z_t) + \sigma(z_t) \rho s_t,$$

while

$$\text{Var}(V(x_{t+1}, s_{t+1}) \mid s_t, \rho) = \sigma(z_t)^2 (1 - \rho^2).$$

Thus a larger ρ makes the signal more informative in two senses. It makes the conditional mean move more strongly with the news s_t , and it reduces the amount of residual uncertainty left for the next period. The same parameter therefore governs both how much the news says and how much uncertainty it resolves.

Now define the dynastic continuation payoff $Y_{t+1} = \Phi(n_t) V(x_{t+1}, s_{t+1})$. Under (21), Y_{t+1} is conditionally normal. The CARA certainty equivalent is therefore

$$K(n_t, z_t; \rho, s_t) = \Phi(n_t) [\mu(z_t) + \sigma(z_t) \rho s_t] - \frac{\gamma}{2} \Phi(n_t)^2 \sigma(z_t)^2 (1 - \rho^2). \quad (22)$$

The first term is the conditional mean value of descendants. The second term is the risk penalty. This penalty scales with $\Phi(n_t)^2$ rather than $\Phi(n_t)$ because the shock is common across children. A larger family is not a diversified portfolio of independent child outcomes. It is a larger exposure to the same future environment.

Using (22), the parent’s reduced problem is

$$\max_{n, z} \left\{ u(m_t - n(q_t + z)) + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K(n, z; \rho, s_t) \right\}. \quad (23)$$

This reduced problem keeps the central recursive logic. Fertility and investment are chosen today, the continuation value is evaluated pessimistically over signal informativeness, and risk

sensitivity penalizes unresolved common risk.

Remark 1 (The two-period example with an exogenous child payoff). *The reduced problem (23) contains a simple two-period model as a special case. Fix investment at $\bar{z}$, define*

$$\mu \equiv \mu(\bar{z}), \quad \sigma \equiv \sigma(\bar{z}), \quad q \equiv q_t + \bar{z},$$

and let

$$X_{t+1} = \mu + \sigma \left[\rho s_t + \sqrt{1 - \rho^2} \varepsilon_{t+1} \right]$$

be the consumption-equivalent payoff from children. This payoff can be interpreted as the child's future well-being, old-age support, family-network value, and any future costs or benefits that move with the common environment.

The parent's problem becomes

$$V(m_t, s_t) = \max_{n \geq 0} \left\{ u(m_t - qn) + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \left[\Phi(n)(\mu + \sigma \rho s_t) - \frac{\gamma}{2} \Phi(n)^2 \sigma^2 (1 - \rho^2) \right] \right\}.$$

This is a static Barro–Becker fertility problem with ambiguous public information. The closed-form analysis below uses this special case, but the interpretation is recursive: X_{t+1} is a tractable stand-in for the next generation's endogenous continuation value.

3.2 Benchmark: known signal informativeness

We first remove ambiguity about the quality of the signal. Suppose the parent knows the signal-informativeness parameter ρ . Then there is no inner minimization over ρ . The parent takes the signal structure as given and chooses consumption, fertility, and investment.

Under the local normal representation, the continuation term is (22). For an interior optimum, with

$$c_t = m_t - n_t(q_t + z_t),$$

the first-order conditions are

$$u'(c_t)(q_t + z_t) = \beta \Phi'(n_t) \left[\mu(z_t) + \sigma(z_t) \rho s_t - \gamma \Phi(n_t) \sigma(z_t)^2 (1 - \rho^2) \right], \quad (24)$$

$$n_t u'(c_t) = \beta \Phi(n_t) \left[\mu_z(z_t) + \sigma_z(z_t) \rho s_t - \gamma \Phi(n_t) \sigma(z_t) \sigma_z(z_t) (1 - \rho^2) \right]. \quad (25)$$

The useful benchmark question is how fertility changes when the signal becomes more informative, holding the realized news s_t fixed. The answer is not monotone. More informative bad news lowers the conditional mean, but it also reduces residual uncertainty. The following result

makes this tradeoff explicit.

Proposition 3 (Known signal informativeness). *Fix $z_t = z$ and suppose the fertility problem has a unique interior solution $n^*(\rho, s_t)$ satisfying the second-order condition. Assume $\Phi'(n^*) > 0$ and $\sigma(z) > 0$. Under the local normal representation (21),*

$$\text{sgn} \left(\frac{\partial n^*}{\partial \rho} \right) = \text{sgn} [s_t + 2\gamma\Phi(n^*)\sigma(z)\rho]. \quad (26)$$

Thus higher signal informativeness raises fertility if and only if

$$s_t > -2\gamma\Phi(n^*)\sigma(z)\rho.$$

With endogenous z_t , let H be the Hessian of the objective with respect to (n, z) , evaluated at the optimum, and suppose H is negative definite. Then

$$\begin{pmatrix} \frac{\partial n^*}{\partial \rho} \\ \frac{\partial z^*}{\partial \rho} \end{pmatrix} = -H^{-1} \begin{pmatrix} \beta\Phi'(n^*)\sigma(z^*)[s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho] \\ \beta\Phi(n^*)\sigma_z(z^*)[s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho] \end{pmatrix}. \quad (27)$$

Proof. The proof is in Appendix A.4. □

The sign condition separates the mean channel from the risk-resolution channel. If the news is good, $s_t > 0$, higher informativeness raises the conditional mean and also reduces residual risk. Both effects increase the risk-adjusted value of children, so fertility rises with ρ . If the news is bad, $s_t < 0$, the two channels move in opposite directions. A more informative signal makes the bad news more relevant, which lowers the conditional mean. But it also reduces uncertainty, which lowers the risk penalty. For moderately bad news,

$$-2\gamma\Phi(n^*)\sigma(z)\rho < s_t < 0,$$

the risk-resolution channel dominates, so fertility still rises with informativeness. For sufficiently bad news,

$$s_t < -2\gamma\Phi(n^*)\sigma(z)\rho,$$

the mean channel dominates, and informativeness lowers fertility.

With endogenous investment, (27) shows that the same mean-versus-risk tradeoff affects

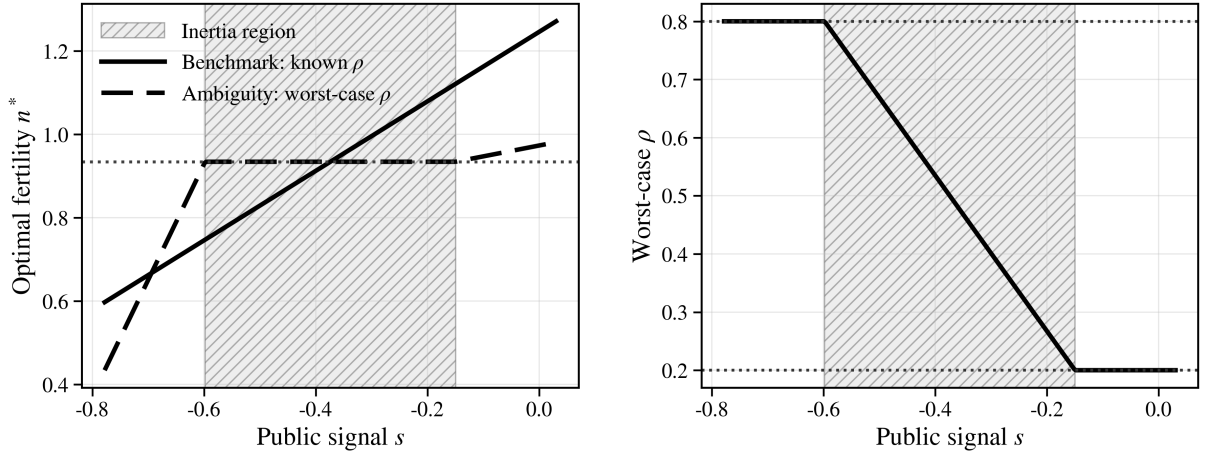


FIGURE 1. Optimal fertility under known and ambiguous signal informativeness.

Notes: The figure compares optimal fertility when signal informativeness is known with optimal fertility when the parent evaluates the signal under the worst admissible informativeness. Parameters are $u(c) = c$, $\Phi(n) = n$, $y = 10$, $q = 1$, $\beta = 0.95$, $\mu = 1.8$, $\sigma = 1$, $\gamma = 0.8$, $\underline{\rho} = 0.2$, and $\bar{\rho} = 0.8$. The known- ρ benchmark uses $\rho = 0.5$. Under these values, the inertia fertility level is $n^0 = 0.934$, and the inertia region is approximately $s \in [-0.598, -0.149]$.

both margins. The common scalar factor

$$s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho$$

appears in the response of both first-order conditions to ρ . When this factor is zero, a marginal change in known informativeness has no first-order effect on either margin. This common structure is useful for the ambiguity analysis below. The worst-case belief will choose informativeness by trading off the same mean and risk-resolution forces, but now from the parent's least favorable perspective.

Figure 1 shows the central mechanism. The left panel plots optimal fertility as the public signal s changes. The solid line is the benchmark in which signal informativeness is known. In that case, the parent treats the signal with a fixed ρ , so news passes directly into the posterior value of children. As the signal becomes less negative, the risk-adjusted value of children rises, and fertility rises smoothly. The dashed line shows the ambiguity case. The parent does not take ρ as fixed. Instead, each fertility choice is evaluated under the least favorable admissible interpretation of the signal. For very bad signals, the worst case sets $\rho_t^* = \bar{\rho}$: the parent acts as if bad news is maximally informative. Fertility is therefore low. As the signal improves, fertility initially rises.

The distinctive region is the shaded interval. In this region, fertility is flat even though the public signal is changing. The reason is shown in the right panel. As the signal becomes

less bad, the worst-case belief adjusts by lowering ρ_t^* . The parent is no longer made worse off mainly by treating bad news as highly informative. Instead, the worst-case belief shifts toward a less informative signal, which leaves more residual uncertainty. Thus the adversarial belief substitutes between two forms of pessimism: making bad news more relevant and leaving more risk unresolved. Inside the shaded region these two forces exactly offset in the fertility margin, so the optimal number of children remains unchanged. Once ρ_t^* reaches the lower bound $\underline{\rho}$, the worst-case belief can no longer reduce informativeness further. From that point on, additional improvements in the signal again pass through to the risk-adjusted value of children, and fertility starts to move. Thus the flat region is bounded on the left by the point where the worst-case belief leaves $\bar{\rho}$, and on the right by the point where it reaches $\underline{\rho}$.

This is why the result is an inertia result. Ambiguity does not simply lower fertility at every signal. Instead, it changes the sensitivity of fertility to news. In the inertia region, the parent does not ignore the signal. The worst-case interpretation of the signal changes in exactly the way needed to keep the marginal value of fertility unchanged. Public news affects the worst-case belief, but not the fertility choice.

3.3 Ambiguous signal informativeness

We now restore ambiguity about the informativeness of the public signal. For a given family plan (n, z) , the parent evaluates continuation value under the worst admissible value of ρ , or $\min_{\rho \in [\underline{\rho}, \bar{\rho}]} K(n, z; \rho, s_t)$. Using (22), the derivative with respect to ρ is

$$K_\rho(n, z; \rho, s_t) = \Phi(n)\sigma(z)s_t + \gamma\Phi(n)^2\sigma(z)^2\rho,$$

and

$$K_{\rho\rho}(n, z; \rho, s_t) = \gamma\Phi(n)^2\sigma(z)^2 > 0.$$

Thus K is strictly convex in ρ . The worst-case ρ is the unconstrained minimizer, projected back onto the ambiguity interval. Hence

$$\rho^w(n, z, s_t) = \text{Proj}_{[\underline{\rho}, \bar{\rho}]} \left(-\frac{s_t}{\gamma\Phi(n)\sigma(z)} \right). \quad (28)$$

This shows how pessimism works. Assume $0 \leq \underline{\rho} < \bar{\rho}$. If the news is good, $s_t \geq 0$, the unconstrained minimizer is nonpositive. The worst case therefore chooses the lower bound, $\rho^w = \underline{\rho}$. Good news is treated as weakly informative. If the news is very bad ($s_t \leq -\gamma\Phi(n)\sigma(z)\bar{\rho}$), the projection binds at the upper bound, so $\rho^w = \bar{\rho}$. Very bad news is treated as highly informative.

Between these two cases, for

$$-\gamma\Phi(n)\sigma(z)\bar{\rho} < s_t < -\gamma\Phi(n)\sigma(z)\underline{\rho},$$

the worst case is interior so that

$$\rho^w(n, z, s_t) = -\frac{s_t}{\gamma\Phi(n)\sigma(z)}.$$

This is the key region. The worst-case belief moves with the signal and with the parent's own plan. A larger planned family, or an investment choice that raises exposure $\sigma(z)$, changes which interpretation of the news is most harmful. The asymmetry is the signature of ambiguity aversion. Favorable news is discounted, while unfavorable news is taken seriously. But the main result comes from the interior region, where the worst-case belief can adjust smoothly.

Proposition 4 (Fertility information inertia). *Let (n^0, z^0) be the unique interior maximizer of the signal-free problem*

$$\max_{n,z} \left\{ u(m_t - n(q_t + z)) + \beta \left[\Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 \right] \right\}. \quad (29)$$

Suppose the objective in (29) is strictly concave in a neighborhood of (n^0, z^0) , with $\Phi(n^0) > 0$ and $\sigma(z^0) > 0$. If the signal satisfies

$$-\gamma\Phi(n^0)\sigma(z^0)\bar{\rho} < s_t < -\gamma\Phi(n^0)\sigma(z^0)\underline{\rho}, \quad (30)$$

then the worst-case signal informativeness is interior at (n^0, z^0) , and the optimal policies in (23) satisfy, locally,

$$n^*(s_t) = n^0, \quad z^*(s_t) = z^0, \quad c^*(s_t) = m_t - n^0(q_t + z^0). \quad (31)$$

Thus, throughout the inertia region, optimal fertility, child investment, and consumption are locally independent of the public signal.

If the objective in (29) is globally strictly concave on the feasible set, and if the worst-case belief remains interior on the relevant choice set, then (31) is the unique global optimum. The boundary points of (30) can be included by continuity, but at those points the worst-case belief is on the boundary rather than strictly interior.

Proof. The proof is available in Appendix A.5. □

3.4 Fertility inertia

Equation (A.1) is the core of the result. After the worst-case belief is chosen, the minimized continuation value is

$$\Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 - \frac{s_t^2}{2\gamma}.$$

The first two terms are exactly the value the parent would assign if the signal were uninformative: expected dynastic value minus the full risk penalty from unresolved common risk.

The third term is different. It is a welfare cost of ambiguous news, but it is the same for every family plan. It lowers the parent's value, but it does not change the marginal value of fertility or child investment. This is why public news has no local effect on choices inside the inertia region. The parent is not ignoring the news. Instead, the worst-case interpretation of the news adjusts in exactly the way needed to make the signal irrelevant at the margin. Bad news lowers the mean continuation value, but a lower worst-case ρ leaves more residual uncertainty. In the interior region, the adversarial belief balances these two forces for every nearby plan.

Inside the inertia region, the first-order conditions are therefore

$$u'(c^0)(q_t + z^0) = \beta\Phi'(n^0) \left[\mu(z^0) - \gamma\Phi(n^0)\sigma(z^0)^2 \right], \quad (32)$$

$$n^0 u'(c^0) = \beta\Phi(n^0) \left[\mu_z(z^0) - \gamma\Phi(n^0)\sigma(z^0)\sigma_z(z^0) \right], \quad (33)$$

where

$$c^0 = m_t - n^0(q_t + z^0).$$

Neither condition contains s_t . An observer who only looks at household choices inside this region would see no response to public information, even though information is central to the parent's evaluation problem.

Corollary 1 (Width of the inertia region). *For a fixed signal-free optimum (n^0, z^0) , the length of the fertility-inertia interval is*

$$\gamma \Phi(n^0) \sigma(z^0) (\bar{\rho} - \underline{\rho}). \quad (34)$$

Thus the direct width of the inertia region is larger when dynastic risk sensitivity γ , fertility exposure $\Phi(n^0)$, continuation-value exposure $\sigma(z^0)$, or ambiguity about signal informativeness $\bar{\rho} - \underline{\rho}$ is larger.

The comparative statics have a direct interpretation. A larger γ makes unresolved risk more costly, so the risk-resolution channel is stronger. A larger $\Phi(n^0)$ means the dynasty has more continuation value exposed to the common environment. A larger $\sigma(z^0)$ means the child's

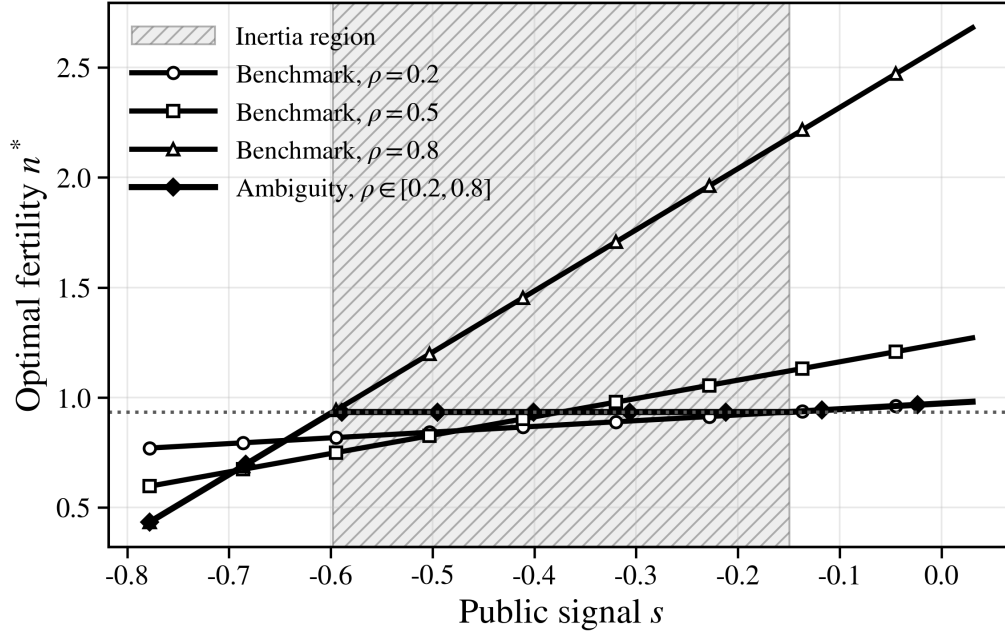


FIGURE 2. Optimal fertility for different values of signal informativeness

Notes: The figure plots optimal fertility as a function of the public signal s . The benchmark curves fix signal informativeness at $\rho = 0.2$, $\rho = 0.5$, and $\rho = 0.8$. The ambiguity case allows $\rho \in [0.2, 0.8]$ and uses the worst-case value of ρ . Parameter values are $u(c) = c$, $\Phi(n) = n$, $y = 10$, $q = 1$, $\beta = 0.95$, $\mu = 1.8$, $\sigma = 1$, and $\gamma = 0.8$. The shaded region is the fertility inertia region. No realizations of ε_{t+1} are simulated; the choices are computed from the certainty equivalent.

continuation value is more sensitive to that environment. A wider ambiguity interval gives the worst-case belief more room to adjust. Each force widens the range of bad signals over which the parent keeps the same family plan. If a change in γ , Φ , or σ also changes the signal-free optimum (n^0, z^0) , then the total comparative static must also include the induced changes in n^0 and z^0 .

The inertia region lies on the bad-news side. Good news is evaluated with the lower-bound belief $\underline{\rho}$, so it is treated as only weakly informative. Very bad news is evaluated with the upper-bound belief $\bar{\rho}$, so it is taken at full worst-case force. Between those two cases, the worst-case belief is interior, and that is where fertility, investment, and consumption become locally flat in the signal.

Figure 2 shows how ambiguity changes the way fertility responds to public news. The three benchmark curves fix signal informativeness at $\rho = 0.2$, $\rho = 0.5$, and $\rho = 0.8$. In each benchmark, fertility rises with the signal: better news raises the risk-adjusted value of children. The slope is steeper when ρ is larger because a more informative signal makes the posterior value of children move more strongly with the news.

The ambiguity curve behaves differently. It is not the response to any single fixed value of ρ . Instead, the parent evaluates each signal under the least favorable admissible interpretation,

$\rho \in [0.2, 0.8]$. For very bad news, the worst case is $\rho = \bar{\rho} = 0.8$. The parent treats bad news as highly informative, so fertility is low and the ambiguity curve follows the high- ρ benchmark. For favorable or only mildly bad news, the worst case is $\rho = \underline{\rho} = 0.2$. The parent treats good news as only weakly informative, so fertility responds only mildly.

The shaded region is the main result. In this interval, the worst-case belief is interior. Instead, it moves with the signal. As the public signal becomes less negative, the adversarial belief lowers ρ , treating the news as less informative but leaving more residual uncertainty. This adjustment keeps the marginal value of fertility unchanged. As a result, optimal fertility is flat at the signal-free level n^0 , even though the public signal is changing.

The figure therefore separates three regimes. For sufficiently bad news, ambiguity amplifies pessimism. The parent acts as if the bad signal is maximally informative. For moderately bad news, ambiguity creates inertia. The worst-case interpretation adjusts, and fertility does not move. For good news, ambiguity discounts optimism. The parent acts as if the favorable signal is minimally informative.

This is not simply a result that ambiguity lowers fertility. The ambiguity curve is piecewise shaped by the worst-case interpretation of the signal. Sometimes it coincides with the high- ρ benchmark, sometimes with the low- ρ benchmark, and in the middle it is flat. The important implication is about sensitivity, not just levels. Ambiguity can make fertility locally unresponsive to news.

The intuition is that the parent faces two kinds of bad interpretations. If ρ is high, bad news strongly lowers the expected value of children, but it also resolves more uncertainty. If ρ is low, the news matters less for the mean, but more risk remains unresolved. Inside the shaded interval, the worst-case belief balances these two forces exactly. The news still affects welfare, but it does not affect the marginal tradeoff that determines fertility. Thus Figure 2 illustrates the paper's inertia mechanism visually. Public information changes the parent's pessimistic belief before it changes the parent's fertility choice. The household is not ignoring news. Rather, ambiguity about what the news means absorbs the news at the belief margin, leaving the family plan unchanged over a range of signals.

3.5 Closed-form special case and quantitative illustration

We now use the simplest version of the two-period model to obtain a closed-form policy rule. Let $u(c) = c$, $\Phi(n) = n$, and fix child investment at $z = 0$, then

$$\mu \equiv \mu(0), \quad \sigma \equiv \sigma(0), \quad q_t = q, \quad y \equiv m_t.$$

Thus q is the resource cost per child and y is current resources.

Starting with the signal-free problem, the parent chooses fertility subject to feasibility, $0 \leq n \leq y/q$:

$$\max_{0 \leq n \leq y/q} \left\{ y - qn + \beta \left[n\mu - \frac{\gamma}{2} n^2 \sigma^2 \right] \right\}.$$

The objective is strictly concave because the quadratic term in n is negative

$$\frac{\partial^2}{\partial n^2} \left\{ y - qn + \beta \left[n\mu - \frac{\gamma}{2} n^2 \sigma^2 \right] \right\} = -\beta\gamma\sigma^2 < 0.$$

Hence signal-free fertility from the interior first-order condition is

$$n^0 = \frac{\mu - q/\beta}{\gamma\sigma^2}. \quad (35)$$

This solution is interior when

$$\mu > q/\beta \quad \text{and} \quad qn^0 < y.$$

The first condition says that the mean dynastic payoff from a child must exceed the discount-adjusted cost of that child. The second condition says that the implied fertility choice is affordable.

Define

$$\lambda \equiv \frac{\mu - q/\beta}{\sigma}.$$

This is a reproductive Sharpe ratio. It measures the excess mean payoff from parenthood per unit of common parenthood risk. Using (35),

$$\gamma\sigma n^0 = \lambda.$$

Therefore the inertia interval becomes

$$s_t \in [-\lambda\bar{\rho}, -\lambda\underline{\rho}], \quad \text{with width} \quad \lambda(\bar{\rho} - \underline{\rho}). \quad (36)$$

This formula gives a simple interpretation of the no-response region. Its width is the product of two forces. The first is λ , the risk-adjusted attractiveness of parenthood. The second is $\bar{\rho} - \underline{\rho}$, the amount of ambiguity about signal informativeness. A wider ambiguity set gives the worst-case belief more room to reinterpret the news. A higher λ means the parent chooses a larger family position, so more dynastic value is exposed to the common environment. As a result,

ambiguous news is able to freeze fertility over a wider range of signals.

Remark 2 (The full policy function). *The same linear specification also gives closed forms outside the inertia region. Suppose the worst-case belief is fixed at a boundary value $\hat{\rho} \in \{\underline{\rho}, \bar{\rho}\}$.*

Then the parent's objective is

$$y - qn + \beta \left[n(\mu + \sigma \hat{\rho} s_t) - \frac{\gamma}{2} n^2 \sigma^2 (1 - \hat{\rho}^2) \right].$$

The interior first-order condition gives

$$n^*(s_t; \hat{\rho}) = \frac{\mu + \sigma \hat{\rho} s_t - q/\beta}{\gamma \sigma^2 (1 - \hat{\rho}^2)}.$$

Thus, as long as the solution remains interior and $0 \leq \underline{\rho} < \bar{\rho} < 1$, the ambiguous-signal policy is

$$n^*(s_t) = \begin{cases} \frac{\mu + \sigma \bar{\rho} s_t - q/\beta}{\gamma \sigma^2 (1 - \bar{\rho}^2)}, & s_t \leq -\lambda \bar{\rho}, \\ n^0, & -\lambda \bar{\rho} \leq s_t \leq -\lambda \underline{\rho}, \\ \frac{\mu + \sigma \underline{\rho} s_t - q/\beta}{\gamma \sigma^2 (1 - \underline{\rho}^2)}, & s_t \geq -\lambda \underline{\rho}. \end{cases} \quad (37)$$

The three pieces paste together continuously. At $s_t = -\lambda \bar{\rho}$, the upper-bound segment equals n^0 . At $s_t = -\lambda \underline{\rho}$, the lower-bound segment also equals n^0 .

The policy is therefore continuous but kinked. It is flat in the inertia region, where the worst-case belief is interior. Outside that region, its slope is

$$\frac{\partial n^*(s_t; \hat{\rho})}{\partial s_t} = \frac{\hat{\rho}}{\gamma \sigma (1 - \hat{\rho}^2)}.$$

Since $\rho/(1 - \rho^2)$ is increasing on $[0, 1)$, the very-bad-news segment, governed by $\bar{\rho}$, is steeper than the good-news segment, governed by $\underline{\rho}$. Ambiguity therefore produces an asymmetric response to news: sufficiently bad news moves fertility sharply, while good news is discounted and moves fertility only weakly.

Figure 3 varies the size of the ambiguity set while keeping its midpoint fixed. This isolates ambiguity itself from average signal informativeness. When $\Delta_\rho = 0$, there is no ambiguity and no flat segment. Hence, fertility responds smoothly to the signal. As Δ_ρ increases, the flat segment expands. The reason is exactly the width formula $\lambda(\bar{\rho} - \underline{\rho})$. A wider ambiguity set gives the worst-case belief more room to move. For moderately bad news, the worst-case belief can adjust

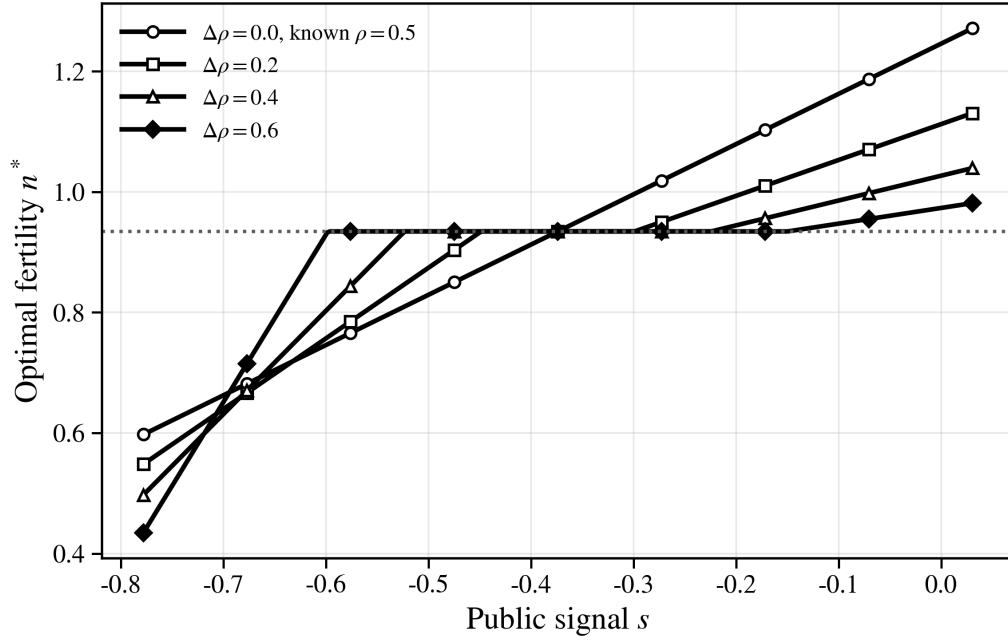


FIGURE 3. Fertility inaction regions for different ambiguity sizes

Notes: The figure plots optimal fertility as a function of the public signal s for different ambiguity-set widths $\Delta\rho = \bar{\rho} - \underline{\rho}$. The midpoint of the ambiguity set is held fixed at $(\underline{\rho} + \bar{\rho})/2 = 0.5$. Thus larger $\Delta\rho$ means more ambiguity about signal informativeness, holding average signal informativeness fixed. The case $\Delta\rho = 0$ is the known- ρ benchmark with $\rho = 0.5$. The other curves use ambiguity intervals $[0.4, 0.6]$, $[0.3, 0.7]$, and $[0.2, 0.8]$. The flat portions are fertility inaction regions. In this closed-form case, their width is $\lambda_F(\bar{\rho} - \underline{\rho}) = \gamma\sigma n^0(\bar{\rho} - \underline{\rho})$. Parameter values are $u(c) = c$, $\Phi(n) = n$, $y = 10$, $q = 1$, $\beta = 0.95$, $\mu = 1.8$, $\sigma = 1$, and $\gamma = 0.8$.

informativeness instead of forcing the parent to adjust fertility. The larger the ambiguity interval, the larger the set of signals over which this belief adjustment absorbs the news.

Because the midpoint is fixed, the center of the inertia region is also fixed

$$-\lambda \left(\frac{\underline{\rho} + \bar{\rho}}{2} \right).$$

Increasing ambiguity therefore expands the no-response region around the same average interpretation of the signal. The figure shows that it is not low informativeness by itself that generates inertia. It is uncertainty about informativeness.

3.6 Economic implications

The mechanism can be summarized as follows. Public news about parenthood has two effects. It changes the expected value of having children, and it changes how much uncertainty remains. Bad news lowers the conditional mean value of children, but if the news is informative, it also resolves

uncertainty. A cautious parent who is unsure how informative the news is evaluates the signal under the worst admissible interpretation. In the inertia region, that worst-case interpretation balances the mean effect and the risk-resolution effect so that the news drops out of the marginal conditions for fertility, child investment, and consumption.

The signal still matters for welfare. In the closed-form case, the minimized continuation value contains the term

$$-\frac{s_t^2}{2\gamma}.$$

This term lowers the parent's value when the signal is nonzero. But it is the same for every fertility choice. It is therefore a level effect, not a marginal effect. The parent is worse off after ambiguous bad news, but the optimal family plan does not change.

This distinction is important empirically. If fertility does not respond to vague public news, one might conclude that households are inattentive, uninformed, or subject to large adjustment costs. The model gives a different explanation. Households may process the news fully, but remain uncertain about how strongly it applies to their own future. Their caution shows up as a changing worst-case interpretation of the signal, not as an immediate change in fertility.

The mechanism also differs from standard adjustment costs. Adjustment costs usually make responses sluggish in a roughly symmetric way. Here the response is asymmetric. Good news is interpreted through the lower bound $\underline{\rho}$ and therefore has a muted effect. Very bad news is interpreted through the upper bound $\bar{\rho}$ and therefore has a much sharper effect. Between the two lies the inertia region, where fertility does not move at all.

The policy implication is that pronatalist policy should distinguish between generosity and legibility. A policy that raises average generosity raises μ or lowers q . This can raise desired fertility and may also change the width of the inertia region indirectly by changing the signal-free optimum n^0 . But it does not by itself remove ambiguity about how public conditions map into a household's own future parenthood payoff.

A policy that makes parenthood payoffs more legible works differently. It reduces the ambiguity interval $\bar{\rho} - \underline{\rho}$. That directly shrinks the range of signals over which households remain inert. Examples include guaranteed childcare slots rather than lotteries, transparent subsidy eligibility rules, stable long-horizon benefit commitments, enforceable parental-leave and job-return rights, caps on child health or education costs, and housing support whose eligibility does not depend on uncertain future reassessments.

The model therefore predicts that vague transfers may have limited short-run effects when households are inside the inertia region. More specific and credible policies can have larger effects because they do not merely increase the average payoff from children. They reduce uncertainty

about what that payoff will be. In the language of the model, they shrink $\bar{\rho} - \underline{\rho}$ and enable decisions that were previously stagnated in the no-response region.

4 Conclusion

This paper develops a theory of fertility choice when parents are unsure not only about the future, but also about how informative current public signals are for their own descendants. The novel contribution is to introduce an insight of Knightian uncertainty into the canonical Barro–Becker framework (Becker and Barro 1988; Barro and Becker 1989). Existing dynastic fertility models assume parents know the mapping from observed news to their children’s future states. We relax this assumption by letting the signal-informativeness parameter be ambiguous, so that after observing public news, a parent evaluates each plan under the least favorable admissible interpretation of how that news applies to her descendants.

The new insight that emerges is informational fertility inertia. Because the worst-case interpretation of a signal depends on its content, ambiguity produces an asymmetric, endogenous response to news. Favorable information is treated as weakly informative, severely unfavorable information as highly informative, and moderately bad news yields an interior worst case that moves continuously with the signal and the household’s own plan. The result is a region in which fertility choices remain unresponsive to public news that is vague or hard to interpret. This is a new prediction, distinct from standard risk-based explanations of fertility behavior. Two channels the model isolates drive it. Through the mean channel, greater informativeness makes continuation value respond more sharply to news. Through the risk-resolution channel, it resolves uncertainty and lowers the residual risk a parent faces.

The framework also reframes how pronatalist policy should be evaluated. The standard lens treats policy as transfers that raise the expected payoff from children or lower their cost. Our model identifies a separate and previously overlooked margin. The legibility of how public conditions map to private outcomes. Policies such as clear eligibility rules, guaranteed childcare, stable leave rights, and credible long-horizon commitments work not by raising average payoffs but by shrinking the ambiguity interval and narrowing the inertia region. This distinction between policies that make children more rewarding and policies that make their rewards more interpretable is the model’s central practical insight. It can explain why generous but opaque and general programs may fail to incentive households.

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Appendices

A Proofs

A.1 Derivation of CARA

Given the future net payoff of a marginal child X_{t+1} , the conditional mean

$$\mathbb{E}[X_{t+1}|s, \rho] = \mu + \sigma \rho s$$

and

$$\text{Var}(X_{t+1}|s, \rho) = \sigma^2(1 - \rho^2).$$

The total payoff of the parent having n children is

$$Z_{t+1} = \Phi(n)X_{t+1}$$

where $\Phi(n)$ is the effective dynastic discounting factor. Conditional on signal s and parameter ρ ,

$$Z_{t+1}|s, \rho \sim N(m, v)$$

where

$$m = \Phi(n)[\mu + \sigma \rho s]$$

and

$$v = \Phi(n)^2 \sigma^2(1 - \rho^2).$$

Given that the utility function is CARA, the parent evaluates the risky future dynastic payoff Z ,

$$U(Z) = -\exp(-\gamma Z)$$

where $\gamma > 0$ is absolute risk aversion. Let

$$-\exp(-\gamma CE) = \mathbb{E}(-\exp(-\gamma Z))$$

and using $Z \sim N(m, v)$, we get

$$\mathbb{E}[\exp(-\gamma Z)] = \exp\left(-\gamma m + \frac{1}{2}\gamma^2 v\right).$$

Therefore,

$$\exp(-\gamma CE) = \exp\left(-\gamma m + \frac{1}{2}\gamma^2 v\right)$$

which gives

$$CE = m - \frac{\gamma}{2}v = \mathbb{E}[Z] - \frac{\gamma}{2}\text{Var}(Z).$$

Substituting the conditional mean and variance of random variable Z yields the equation

$$CE(n; s, \rho) = \Phi(n)[\mu + \sigma\rho s] - \frac{\gamma}{2}\Phi(n)^2\sigma^2(1 - \rho^2).$$

A.2 Proof of Proposition 1

Proof. Let $C_b(\mathcal{X} \times \mathcal{S})$ be the space of bounded continuous functions on the state space, with the sup norm $\|\cdot\|_\infty$, a complete metric space. Write a feasible action as $a = (c, n, z) \in \Gamma(x, s)$ and $V' = V(x', s')$. We show T maps C_b into itself and is a contraction with modulus $\beta\bar{\Phi}$, then characterize the fixed point by measurable selections.

We use two properties of $\mathcal{C}_{\gamma, \rho, z}$, both immediate from $\mathcal{C}_{\gamma, \rho, z}[Y] = -\gamma^{-1} \log \mathbb{E}_{\rho, z}[e^{-\gamma Y}]$ (and from the conditional expectation when $\gamma = 0$): monotonicity, $Y_1 \leq Y_2 \Rightarrow \mathcal{C}_{\gamma, \rho, z}[Y_1] \leq \mathcal{C}_{\gamma, \rho, z}[Y_2]$; and translation invariance, $\mathcal{C}_{\gamma, \rho, z}[Y + k] = \mathcal{C}_{\gamma, \rho, z}[Y] + k$ for constant k . Together they give the sup-norm bound: if $\|Y_1 - Y_2\|_\infty \leq \eta$ then

$$|\mathcal{C}_{\gamma, \rho, z}[Y_1] - \mathcal{C}_{\gamma, \rho, z}[Y_2]| \leq \eta. \quad (*)$$

T maps C_b into C_b . Fix $V \in C_b$. By Assumption 1, Γ is nonempty, compact-valued, and continuous, $[\underline{\rho}, \bar{\rho}]$ is compact, u is continuous and bounded on the graph of Γ ($|u| \leq \bar{u}$), and $0 \leq \Phi(n) \leq \bar{\Phi}$. Then $|\Phi(n)V'| \leq \bar{\Phi}\|V\|_\infty$, and applying $(*)$ against the constant bounds gives $|\mathcal{C}_{\gamma, \rho, z}[\Phi(n)V']| \leq \bar{\Phi}\|V\|_\infty$ uniformly in ρ , hence $|(TV)(x, s)| \leq \bar{u} + \beta\bar{\Phi}\|V\|_\infty$; TV is bounded. For continuity, set $H_V(x, s, a, \rho) = u(x, s, a) + \beta\mathcal{C}_{\gamma, \rho, z}[\Phi(n)V']$. Under Assumption 1 the transition law is continuous and the integrand $e^{-\gamma\Phi(n)V'}$ is bounded and continuous, so by bounded convergence H_V is continuous in (x, s, a, ρ) . Since $[\underline{\rho}, \bar{\rho}]$ is compact and fixed in (x, s, a) , Berge's theorem makes $G_V(x, s, a) := \min_\rho H_V(x, s, a, \rho)$ continuous with nonempty compact-valued argmin; applying Berge's theorem again with the compact-valued continuous Γ makes $(TV)(x, s) = \max_{a \in \Gamma(x, s)} G_V(x, s, a)$ continuous.

T is a contraction. Let $V_1, V_2 \in C_b$ and $\Delta := \|V_1 - V_2\|_\infty$. For fixed feasible (x, s, a, ρ) , $|V'_1 - V'_2| \leq \Delta$ and $\Phi(n) \geq 0$ give $\|\Phi(n)V'_1 - \Phi(n)V'_2\|_\infty \leq \bar{\Phi}\Delta$, so by $(*)$,

$$|\mathcal{C}_{\gamma, \rho, z}[\Phi(n)V'_1] - \mathcal{C}_{\gamma, \rho, z}[\Phi(n)V'_2]| \leq \bar{\Phi}\Delta$$

uniformly in ρ . The minimization over ρ and the maximization over a are each nonexpansive in the sup norm: if $\sup_\rho |f_\rho - g_\rho| \leq \eta$ then $|\min_\rho f_\rho - \min_\rho g_\rho| \leq \eta$, and likewise for $\max$ (from $f \leq g + \eta$, take the relevant extremum of both sides, then swap roles). The payoff $u(x, s, a)$ cancels in the difference. Passing the bound through the inner min, multiplying by β , and passing through the outer max gives, for every (x, s) ,

$$|(TV_1)(x, s) - (TV_2)(x, s)| \leq \beta\bar{\Phi}\Delta,$$

hence $\|TV_1 - TV_2\|_\infty \leq \beta\bar{\Phi}\|V_1 - V_2\|_\infty$. Since $\beta\bar{\Phi} < 1$ by Assumption 1, T is a contraction on C_b , and Banach's fixed-point theorem gives a unique $V \in C_b$ with $V = TV$.

Characterization. At this fixed point define

$$\mathcal{R}_V(x, s, a) := \arg \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \mathcal{C}_{\gamma, \rho, z}[\Phi(n)V'], \quad \mathcal{P}_V(x, s) := \arg \max_{a \in \Gamma(x, s)} \left\{ u(x, s, a) + \beta \min_{\rho} \mathcal{C}_{\gamma, \rho, z}[\Phi(n)V'] \right\}.$$

By the continuity and compactness above, Berge's theorem makes both correspondences nonempty and compact-valued, each admitting a measurable selection. For any selections

$$a^*(x, s) = (c^*, n^*, z^*)(x, s) \in \mathcal{P}_V(x, s) \text{ and } \rho^*(x, s) \in \mathcal{R}_V(x, s, a^*(x, s)),$$

$$V(x, s) = u(x, s, a^*(x, s)) + \beta \mathcal{C}_{\gamma, \rho^*(x, s), z^*(x, s)}[\Phi(n^*(x, s))V'] \quad \text{for every } (x, s).$$

The worst-case belief $\rho^*(x, s)$ is the inner minimizer evaluated at the chosen policy $a^*(x, s)$, which is the belief endogeneity used in the inertia analysis. $\square$

A.3 Proof of Proposition 2

Proof. Fix the current state (x_t, s_t) and suppress it where no confusion can arise. Let

$$J(n, z; x_t, s_t) := \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n, z; x_t, s_t).$$

The parent's problem can be written as

$$\max_{c, n, z} \{u(c) + \beta J(n, z; x_t, s_t)\}$$

subject to

$$c + n(q_t + z) \leq m_t.$$

Since $u'(c) > 0$, the budget constraint binds at an optimum. Hence

$$c_t^* = m_t - n_t^*(q_t + z_t^*),$$

which gives (12).

Let λ_t denote the multiplier on the budget constraint. For an interior solution, the first-order conditions are

$$u'(c_t^*) = \lambda_t,$$

$$\beta J_n(n_t^*, z_t^*; x_t, s_t) = \lambda_t(q_t + z_t^*),$$

and

$$\beta J_z(n_t^*, z_t^*; x_t, s_t) = \lambda_t n_t^*.$$

Substituting $\lambda_t = u'(c_t^*)$ gives

$$u'(c_t^*)(q_t + z_t^*) = \beta J_n(n_t^*, z_t^*; x_t, s_t)$$

and

$$n_t^* u'(c_t^*) = \beta J_z(n_t^*, z_t^*; x_t, s_t).$$

It remains to compute the derivatives of J . For fixed ρ , use the differentiable shock representation

$$x_{t+1} = G^x(x_t, s_t, z_t, \varepsilon_{t+1}; \rho), \quad s_{t+1} = G^s(x_t, s_t, z_t, \varepsilon_{t+1}; \rho),$$

where the conditional law of ε_{t+1} depends on ρ but not on n or z . Define

$$Y_{t+1}(n, z, \rho) := \Phi(n)V(G^x(x_t, s_t, z, \varepsilon_{t+1}; \rho), G^s(x_t, s_t, z, \varepsilon_{t+1}; \rho)).$$

For $\gamma > 0$,

$$K_\rho(n, z; x_t, s_t) = -\frac{1}{\gamma} \log \mathbb{E}_\rho[\exp\{-\gamma Y_{t+1}(n, z, \rho)\} \mid x_t, s_t],$$

where the expectation is over the shock ε_{t+1} .

By the maintained differentiability and domination assumptions, differentiation under the expectation is valid. For any control $a \in \{n, z\}$,

$$\begin{aligned} \frac{\partial K_\rho}{\partial a} &= -\frac{1}{\gamma} \frac{\frac{\partial}{\partial a} \mathbb{E}_\rho[\exp\{-\gamma Y_{t+1}\} \mid x_t, s_t]}{\mathbb{E}_\rho[\exp\{-\gamma Y_{t+1}\} \mid x_t, s_t]} \\ &= -\frac{1}{\gamma} \frac{\mathbb{E}_\rho\left[-\gamma \exp\{-\gamma Y_{t+1}\} \frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t\right]}{\mathbb{E}_\rho[\exp\{-\gamma Y_{t+1}\} \mid x_t, s_t]} \\ &= \frac{\mathbb{E}_\rho\left[\exp\{-\gamma Y_{t+1}\} \frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t\right]}{\mathbb{E}_\rho[\exp\{-\gamma Y_{t+1}\} \mid x_t, s_t]}. \end{aligned}$$

Define the distorted measure $Q_{\rho, n, z}$ by

$$\frac{dQ_{\rho, n, z}}{dP_{\rho, z}} = \frac{\exp\{-\gamma Y_{t+1}(n, z, \rho)\}}{\mathbb{E}_{\rho, z}[\exp\{-\gamma Y_{t+1}(n, z, \rho)\} \mid x_t, s_t]}.$$

Equivalently, under the shock representation, this is the exponential tilt of the shock law. Therefore

$$\frac{\partial K_\rho}{\partial a} = \mathbb{E}_{Q_{\rho, n, z}} \left[\frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t \right].$$

Since the inner minimizer ρ_t^* is unique, Danskin's envelope theorem implies

$$J_a(n_t^*, z_t^*; x_t, s_t) = \left. \frac{\partial K_\rho(n_t^*, z_t^*; x_t, s_t)}{\partial a} \right|_{\rho=\rho_t^*}, \quad a \in \{n, z\}.$$

Thus there is no additional term involving the derivative of ρ_t^* with respect to n or z .

Now compute the two derivatives of Y_{t+1} . Since the transition does not depend directly on n ,

$$\frac{\partial Y_{t+1}}{\partial n} = \Phi'(n_t^*)V(x_{t+1}, s_{t+1}).$$

Therefore

$$J_n(n_t^*, z_t^*; x_t, s_t) = \Phi'(n_t^*)\mathbb{E}_{Q_t^*}[V(x_{t+1}, s_{t+1}) \mid x_t, s_t],$$

where $Q_t^* = Q_{\rho_t^*, n_t^*, z_t^*}$. Substituting into the first-order condition for n gives

$$u'(c_t^*)(q_t + z_t^*) = \beta\Phi'(n_t^*)\mathbb{E}_{Q_t^*}[V(x_{t+1}, s_{t+1}) \mid x_t, s_t],$$

which is (13).

For z , the chain rule gives

$$\begin{aligned} \frac{\partial Y_{t+1}}{\partial z} &= \Phi(n_t^*)[\nabla_x V(x_{t+1}, s_{t+1}) \cdot G_z^x(x_t, s_t, z_t^*, \varepsilon_{t+1}; \rho_t^*) \\ &\quad + V_s(x_{t+1}, s_{t+1})G_z^s(x_t, s_t, z_t^*, \varepsilon_{t+1}; \rho_t^*)] \\ &= \Phi(n_t^*)D_z V_{t+1}. \end{aligned}$$

Hence

$$J_z(n_t^*, z_t^*; x_t, s_t) = \Phi(n_t^*)\mathbb{E}_{Q_t^*}[D_z V_{t+1} \mid x_t, s_t].$$

Substituting into the first-order condition for z gives

$$n_t^* u'(c_t^*) = \beta\Phi(n_t^*)\mathbb{E}_{Q_t^*}[D_z V_{t+1} \mid x_t, s_t],$$

which is (14).

Finally, by definition of the inner problem,

$$\rho_t^* \in \arg \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t).$$

If the minimizer is interior and K_ρ is differentiable in ρ , then the usual first-order condition for the inner minimization gives

$$\left. \frac{\partial K_\rho(n_t^*, z_t^*; x_t, s_t)}{\partial \rho} \right|_{\rho=\rho_t^*} = 0.$$

If $\rho_t^* = \underline{\rho}$, the right derivative must be weakly nonnegative:

$$\left. \frac{\partial K_\rho(n_t^*, z_t^*; x_t, s_t)}{\partial \rho} \right|_{\rho=\underline{\rho}} \geq 0.$$

If $\rho_t^* = \bar{\rho}$, the left derivative must be weakly nonpositive:

$$\left. \frac{\partial K_\rho(n_t^*, z_t^*; x_t, s_t)}{\partial \rho} \right|_{\rho=\bar{\rho}} \leq 0.$$

This proves the result. □

A.4 Proof of Proposition 3

Proof. Fix $z_t = z$ and write

$$c(n) = m_t - n(q_t + z).$$

The fertility first-order condition is

$$G(n, \rho; s_t) \equiv -(q_t + z)u'(c(n)) + \beta\Phi'(n) \left[\mu(z) + \sigma(z)\rho s_t - \gamma\Phi(n)\sigma(z)^2(1 - \rho^2) \right] = 0.$$

At the interior optimum, the second-order condition implies $G_n(n^*, \rho; s_t) < 0$. Hence the implicit function theorem gives

$$\frac{\partial n^*}{\partial \rho} = -\frac{G_\rho}{G_n}.$$

Differentiating G with respect to ρ gives

$$G_\rho = \beta\Phi'(n^*)\sigma(z) [s_t + 2\gamma\Phi(n^*)\sigma(z)\rho].$$

Since $\beta > 0$, $\Phi'(n^*) > 0$, $\sigma(z) > 0$, and $G_n < 0$, the sign of $\partial n^*/\partial \rho$ is the sign of

$$s_t + 2\gamma\Phi(n^*)\sigma(z)\rho,$$

which proves (26).

For the case with endogenous z , let

$$W(n, z; \rho, s_t) = u(m_t - n(q_t + z)) + \beta K(n, z; \rho, s_t).$$

The first-order conditions are

$$W_n(n^*, z^*; \rho, s_t) = 0, \quad W_z(n^*, z^*; \rho, s_t) = 0.$$

Totally differentiating this system with respect to ρ yields

$$H \begin{pmatrix} \partial n^*/\partial \rho \\ \partial z^*/\partial \rho \end{pmatrix} + \begin{pmatrix} W_{n\rho} \\ W_{z\rho} \end{pmatrix} = 0,$$

where H is the Hessian of W with respect to (n, z) . Direct differentiation gives

$$W_{n\rho} = \beta\Phi'(n^*)\sigma(z^*)[s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho],$$

and

$$W_{z\rho} = \beta\Phi(n^*)\sigma_z(z^*)[s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho].$$

Solving the linear system gives (27). □

A.5 Proof of Proposition 4

Proof. Wherever the worst-case belief is interior, (28) gives

$$\rho^w(n, z, s_t) = -\frac{s_t}{\gamma\Phi(n)\sigma(z)}.$$

Substituting this value into (22) gives

$$\min_{\rho} K(n, z; \rho, s_t) = \Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 - \frac{s_t^2}{2\gamma}. \quad (\text{A.1})$$

The last term depends on the signal, but it does not depend on either choice variable, n or z .

Therefore, on any region of the choice space where the worst-case belief is interior, the parent's problem is the signal-free problem (29) plus the constant $-\beta s_t^2/(2\gamma)$. Adding a constant changes the level of value but not the maximizer.

The signal restriction (30) is equivalent to

$$\underline{\rho} < -\frac{s_t}{\gamma\Phi(n^0)\sigma(z^0)} < \bar{\rho}.$$

Thus the worst-case belief is interior at (n^0, z^0) . By continuity, it remains interior in a neighborhood of (n^0, z^0) . On that neighborhood, the objective is exactly the signal-free objective plus a constant. Since (n^0, z^0) is the unique local maximizer of the signal-free problem, it is also the unique local maximizer of the ambiguous-signal problem. Hence

$$n^*(s_t) = n^0, \quad z^*(s_t) = z^0,$$

and the budget condition gives

$$c^*(s_t) = m_t - n^0(q_t + z^0).$$

If the same interiority condition holds on the relevant feasible set and the signal-free objective is globally strictly concave, the same constant-shift argument applies globally. Therefore (31) is the unique global optimum. □

B Appendix B

This paper is robust to boring bits.